User Manual

for S32 WDG_VR5510 Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	30.06.2023	NXP AASW Team	S32 Real-Time Drivers AUTOSAR 4.4 Version 4.0.2 Release

Chapter 2

Introduction

This User Manual describes NXP Semiconductor AUTOSAR Wdg_VR5510 for S32CC. AUTOSAR Wdg_VR5510 driver configuration parameters and deviations from the specification are described in Driver chapter of this document. AUTOSAR Wdg_VR5510 driver requirements and APIs are described in the AUTOSAR Wdg driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525
- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
C/CPP	C and C++ Source Code
CS	Chip Select
CTU	Cross Trigger Unit
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
FIFO	First In First Out
LSB	Least Signifigant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	Specification of WDG_VR5510 Driver	AUTOSAR Release 4.↩ 4.0
2	S32G2 Reference Manual	Rev. 7, Feb 2023
3	S32G3 Reference Manual	Rev.3, Feb 2023
4	S32G2: Mask Set Errata for Mask 0P77B	Rev. 3.0
5	S32G2: Mask Set Errata for Mask 1P77B	Rev. 1.0
6	S32G3: Mask Set Errata for Mask 1P72B	Rev. 2.1
7	S32G2 Data Sheet	Rev. 6, Dec 2022
8	S32G3 Data Sheet	Rev 2, RC, Feb 2023
9	VR5510 Data Sheet	Rev. 5, April 2022

Chapter 3

Driver

- Requirements
- Driver Design Summary
- Hardware Resources
- Deviations from Requirements
- Driver Limitations
- Driver usage and configuration tips
- Runtime errors
- Symbolic Names Disclaimer

3.1 Requirements

Requirements for this driver are detailed in the Autosar Driver Software Specification document (See Table Reference List).

3.2 Driver Design Summary

The VR5510 is an automotive multi-output power management integrated circuit, with a focus on Gateway, ADAS, V2X, and Infotainment applications. It includes multiple high-efficiency switch modes and linear voltage regulators. It offers external frequency synchronization input and output, for optimized system EMC performance. The VR5510 includes enhanced safety features, with fail-safe output, becoming a full part of a safety-oriented system partitioning, covering both ASIL B and ASIL D safety integrity level. It is developed in compliance with ISO26262 standard. The device could also be configured to operate as a non-safety QM version part. One of these safety features is the Watchdog component of the VR5510 chip. This dedicated watchdog monitoring is implemented to verify and report the correct F-IN frequency range. Different clocks can be sent to F-OUT pin to synchronize an external IC or for diagnostic.

The watchdog is a windowed watchdog for the Simple and the Challenger watchdog. The first half of the window is said CLOSED and the second half is said OPEN. A good watchdog refresh is a good watchdog answer during the OPEN window. A bad watchdog refresh is a bad watchdog answer during the OPEN window, no watchdog refresh during the OPEN window, or a good watchdog answer during the CLOSED window. After a good or a

bad watchdog refresh, a new window period starts immediately for the MCU to keep the synchronization with the windowed watchdog.
INIT_FS must be closed by the first good watchdog refresh before the OTP option timeout (256ms/1024ms/32.↩
5s/67s). Then the watchdog window is running and the MCU must refresh the watchdog in the OPEN window of the watchdog window period. The duration of the watchdog window is configurable from 1ms to 1024ms with the WD_WINDOW [3:0] bits. The new watchdog window is effective after the next watchdog refresh. The watchdog window can be disabled during INIT_FS only. The watchdog disable is effective when the INIT_FS is closed.

3.3 Hardware Resources

The WDG external driver uses the Window Watchdog on VR5510 device.

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR Wdg_VR5510 driver software specification in some places. The table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, not available, not testable or out of scope for the Wdg_VR5510 Driver.
The table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, not available, not testable or out of scope for the Pmic Driver.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, not available, not testable or out of scope for the driver.

Requirement	Status	Description	Notes
SWS_Wdg_00034	N/S	The start address of the watchdog trigger routine shall be statically configurable to a fixed memory location by the user. The user needs to take care thatConfigured memory location is valid for the platform on which driver is being implemented on.This configuration parameter shall only be given if supported/needed by the hardware.	Not applicable for any of the available platforms
SWS_Wdg_00161	N/S	To access the internal watchdog hardware, the corresponding Wdg module instance shall access the hardware for watchdog servicing directly.	The Wdg_VR55XX driver is used with an external watchdog hardware.

Requirement	Status	Description	Notes
SWS_Wdg_00166	N/S	The routine servicing an internal watchdog shall be implemented as an interrupt routine driven by a hardware timer	The Wdg_VR55XX driver is used with an external watchdog hardware.
SWS_Wdg_00172	N/S	If more than one watchdog driver instance exits on an ECU (namely an external and an internal one) the A↩ PI names and instance specific type names specified in this chapter shall be made unique by expansion according to SRS_BSW_00347.	The Wdg_VR55XX driver is used with an external watchdog hardware. It has only 1 instance.
SWS_Wdg_00175	N/S	These requirements are not applicable to this specification.	These requirement are applied for this release,
ECUC_Wdg_00118	N/S	WdgTriggerLocation: Location (memory address) of the watchdog trigger routine	This paramater functionality is replaced by PR-MCAL-3268.wdg.
ECUC_Wdg_00112	N/S	WdgExternalConfiguration: Configuration items for an external watchdog hardware	The Wdg_VR55XX driver is used with an external watchdog hardware.
ECUC_Wdg_00113	N/S	WdgExternalContainerRef:Reference to eithera DioChannelGroup container in case the hardware watchdog is connected via DIO pinsan Spi↩ SequenceConfiguration container in case the watchdog hardware is accessed via SPI	The Wdg_VR55XX driver is used with an external watchdog hardware.

3.5 Driver Limitations

The following limitations currently apply to the Wdg_VR5510 driver:

1. The watchdog is triggered via I2c command, it depends on the I2c speed. When I2c Speed has low speed, the watchdog might not trigger successfully in the watchdog's window open time. To avoid this problem, we recommend that the user should configure I2c with high speed, using a big window period when it is enabled.

2. When PMIC VR5510 device had already NORMAL_FS state(Eg: boot phase) with a window recovery period enabling, and user re-configures the WDW_RECOVERY in the FS_WD_WINDOW register to DISABLE window (in application phase), the I2C_FS_REQ error state is reported in FS_DIAG_SAFETY register and the FS_COM_G flag will be set in the FS_G_FLAG register, the bad watchdog refresh occur. To avoid this issue, user should not reconfigure WDW_RECOVERY to DISABLE(in the application phase). In addition, user can call the Pmic_GotoInitFS() of Pmic driver, switch VR5510 back to INIT_FS state and then initialize normally.

3. Wdg_VR5510_Example_S32G274A_M7 and Wdg_VR5510_Example_S32G399A_M7 have a warning when project is compiled. This happens because the project must add all the files of WDGIf stub while using only the WdgIf_Types.h file.

4. For non AUTOSAR project, user must include the Wdg_VR5510_VR55XX.h and Wdg_43_VR5510_Ip_[VariantName]_PBcfg.c.h for compile project.

3.6 Driver usage and configuration tips

1. Enable Watchdog by using Pmic to configure OTP fail-safe with device burn OTP without a watchdog.

2. If the device enables Watchdog, the user must enable watchdog monitoring in Pmic Configuration as a node PmicOtpConfiguration/PmicOtpConfiguration_0/PmicOtpFailSafeUnitConfiguration/PmicOtpSafetyIOConfiguration/PmicWatchdogEnable and select the watchdog type according to watchdog OTP.

3. Pmic_Init() of the Pmic module must be called before Wdg_43_VR5510_Init API because the Wdg driver accesses the device via Pmic driver API controls I2c connection.

4. Pmic_InitDevice() of the Pmic module must be called after Wdg_43_VR5510_Init because this function allows triggering the watchdog and initializes the Pmic device.

5. Configure the Watchdog routine used for triggering a GPT callback (Wdg_VR5510_Cbk_GptNotification0 must be configured as a notification callback for the GPT channel intended for triggering)

6. If the device was initialized and switched to Normal_FS state, the Wdg driver will ignore the configuration to fail-safe register. After the driver is initialized, the watchdog will refresh with the default watchdog mode configuration period time. Users can use the Wdg_43_VR5510_SetMode to update the configuration period time. However, In the Normal_FS state, the watchdog window period is only updated when a good watchdog refreshes on the OPEN window duration time. A bad watchdog error may occur because of the refreshing on the CLOSE window duration time of the previous watchdog window which has already been executed in Normal_FS. To avoid this problem, the user should set the watchdog window period disabled in the boot phase, and re-configure it in the application phase.

7. If the user sets the disabling watchdog window period by node WdgWindowPeriodEnable, the window period will set value "FS_WD_WINDOW[WD_WINDOW] = DISABLED"

8. Wdg driver supported both Autosar and Non-Autosar. Integrate the Wdg_VR5510 and Wdg_VR5510_Ip components into S32DS IDE for configuration Autosar and Non-Autosar. Refer to examples Wdg_VR5510_HLD_Example_S32G274A_M7, Wdg_VR5510_IP_Example_S32G274A_M7, Wdg_VR5510_HLD_Example_S32G399A_M7, and Wdg_VR5510_IP_Example_S32G399A_M7 to understand how to use them.

9. Wdg driver supported a task notification where the user can check the FS_G_FLAG register value of the VR5510 device after each watchdog refreshing. Users can use the task notification by enabling the node WdgCheckFaultStatusNotification in the configuration tool. Through this task notification, the user can do the reaction according to each status of FS_G_FLAG value.

3.7 Runtime errors

The driver generates the following DET errors at runtime.

Table 3.3 Default Errors (reported by DET)

Function	Error Code	Condition triggering the error
Wdg_43_VR5510_Init	WDG_VR5510_E_INIT_FAILED	Initialized driver fail.
Wdg_43_VR5510_Init	WDG_VR5510_E_DRIVER_STATE	Invalid driver state when driver initialize.
Wdg_43_VR5510_Init	WDG_VR5510_E_PARAM_CONFIG	Invalid input parameter.
Wdg_43_VR5510_SetMode	WDG_VR5510_E_DRIVER_STATE	Invalid driver state when driver switches mode.
Wdg_43_VR5510_SetMode	WDG_VR5510_E_PARAM_MODE	Invalid input mode.
Wdg_43_VR5510_SetTriggerCondition	WDG_VR5510_E_DRIVER_STATE	Invalid driver state when driver sets timeout.
Wdg_43_VR5510_SetTriggerCondition	WDG_VR5510_E_PARAM_TIMEOUT	Invalid input parameter.
Wdg_43_VR5510_GetVersionInfo	WDG_VR5510_E_PARAM_POINTER	Invalid input parameter

The driver generates the following DEM errors at runtime.

Table 3.5 Default Errors (reported by DEM)

Function	Error Code	Condition triggering the error
Wdg_43_VR5510_Init	WDG_E_DISABLE_REJECTED	Initialization failed by disabling the watchdog.
Wdg_43_VR5510_SetMode	WDG_E_DISABLE_REJECTED	Mode switch failed by disabling the watchdog.
Wdg_43_VR5510_Init	WDG_E_MODE_FAILED	Setting a watchdog mode failed during initialization.
Wdg_43_VR5510_SetMode	WDG_E_MODE_FAILED	Setting a watchdog mode failed during mode switch.

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

#define <Mip>Conf_<Container_ShortName>_<Container_ID>

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module Wdg
  - Container WdgDemEventParameterRefs
    - Reference WDG_E_DISABLE_REJECTED
    - Reference WDG_E_MODE_FAILED
  - Container WdgGeneral
    - Parameter WdgDisableDemReportErrorStatus
    - Parameter WdgDevErrorDetect
    - Parameter WdgDisableAllowed
    - Parameter WdgTimeoutDuration
    - Parameter WdgIndex
    - Parameter WdgInitialTimeout
    - Parameter WdgMaxTimeout
    - Parameter WdgTimeoutMethod
    - Parameter WdgRunArea
    - Parameter WdgTriggerLocation
    - Parameter WdgType
    - Parameter WdgVersionInfoApi
    - Parameter WdgEnableUserModeSupport
    - Parameter WdgCheckFaultStatusNotification
    - Reference WdgEcucPartitionRef
  - Container WdgSettingsConfig
    - Parameter WdgDefaultMode
    - Parameter WdgErrorCounterLimit
    - Parameter WdgRefreshCounterLimit
    - Parameter WdgErrorImpact
    - Parameter WdgWindowPeriodEnable
    - Parameter WdgRecoveryWindowPeriod
    - Parameter WdgServiceKeyValue
    - Reference WdgPmicdeviceRef
    - Reference WdgExternalTriggerCounterRef

        * Container WdgExternalConfiguration
            · Reference WdgExternalContainerRef
        * Container WdgSettingsFast
            · Parameter WdgWindowPeriod
            · Parameter WdgClosedWindowDutyCycle
        * Container WdgSettingsSlow
            · Parameter WdgWindowPeriod
            · Parameter WdgClosedWindowDutyCycle
        * Container WdgSettingsOff
            · Parameter WdgWindowPeriod
            · Parameter WdgClosedWindowDutyCycle
    – Container WdgPublishedInformation
        * Parameter WdgTriggerMode
    – Container CommonPublishedInformation
        * Parameter ArReleaseMajorVersion
        * Parameter ArReleaseMinorVersion
        * Parameter ArReleaseRevisionVersion
        * Parameter ModuleId
        * Parameter SwMajorVersion
        * Parameter SwMinorVersion
        * Parameter SwPatchVersion
        * Parameter VendorApiInfix
        * Parameter VendorId

4.1 Module Wdg

Wdg Configuration of the Wdg (Watchdog driver) module

Included containers:

- WdgDemEventParameterRefs
- WdgGeneral
- WdgSettingsConfig
- WdgPublishedInformation
- CommonPublishedInformation

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-LINK-TIME, VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container WdgDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.3 Reference WDG_E_DISABLE_REJECTED

Reference to the DemEventParameter which shall be issued when the error "Initialization or mode switch failed because it would disable the watchdog" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.4 Reference WDG_E_MODE_FAILED

Reference to the DemEventParameter which shall be issued when the error "Setting a watchdog mode failed (during initialization or mode switch)" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.5 Container WdgGeneral

WdgGeneral

All general parameters of the watchdog external driver are collected here.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.6 Parameter WdgDisableDemReportErrorStatus

Enable/Disable Dem error reporting.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.7 Parameter WdgDevErrorDetect

Wdg Development Error Detect

Compile switch to enable / disable development error detection for this module.

True: Development error detection enabled

False: Development error detection disabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.8 Parameter WdgDisableAllowed

Wdg Disable Allowed

Compile switch to allow / forbid disabling the watchdog driver during runtime.

True: Disabling the watchdog driver at runtime is allowed

False:Disabling the watchdog driver at runtime is not allowed

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.9 Parameter WdgTimeoutDuration

This Timeout is used to wait Pmic go to FS_INIT state

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	50000
max	4294967295
min	1

4.10 Parameter WdgIndex

Wdg Instance Index

Represents the watchdog driver's ID for Instance so that it can be referenced by the watchdog interface.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.11 Parameter WdgInitialTimeout

Wdg Initial Timeout

The initial timeout (sec) for the trigger condition to be initialized during Init function. It shall be not larger than WdgMaxTimeout.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.0
max	65.535
min	0.0

4.12 Parameter WdgMaxTimeout

Wdg Max Timeout

The maximum timeout (sec) to which the watchdog external trigger condition can be initialized.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.0
max	65.535
min	0.0

4.13 Parameter WdgTimeoutMethod

WdgTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If SystemTimer or CustomTimer are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_SYSTEM
literals	['OSIF_COUNTER_DUMMY', 'OSIF_COUNTER_SYSTEM', 'OSIF_COU↩ NTER_CUSTOM']

4.14 Parameter WdgRunArea

Wdg Run Area

Represents the watchdog External driver execution area is either from ROM(Flash) or RAM as required with the particular microcontroller.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ROM
literals	['RAM', 'ROM']

4.15 Parameter WdgTriggerLocation

Wdg Trigger Location

Location (memory address) of the watchdog trigger routine.

Note:Not supported by the current hardware.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	AUTOSAR_ECUC

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	NULL_PTR

4.16 Parameter WdgType

Configures the type of watchdog (WD_SELECTION_OTP).

The watchdog can be either a simple watchdog or a challenger watchdog.

The simple watchdog monitoring is based on a single configurable watchdog key.

The challenger watchdog monitoring is based on a 16-bits pseudo-random word

generated by a LFSR, whose seed is configurable.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	CHALLENGER_WATCHDOG
literals	['CHALLENGER_WATCHDOG', 'SIMPLE_WATCHDOG']

4.17 Parameter WdgVersionInfoApi

Wdg VersionInfo Api

Compile switch to enable or disable the version information API.

True: API enabled

False:API disabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.18 Parameter WdgEnableUserModeSupport

Enable/Disable user mode support.

True: Enabled

False: Disabled

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.19 Parameter WdgCheckFaultStatusNotification

This function will be called in the Wdg_VR5510_Cbk_GptNotification0 after each watchdog refresh.

If this notification is enabled, the driver will use this function to receive the PMIC FS_G_FLAG register value and handle the error report.

The content of notification depends on the user application. User should not integrate too many thing in this, because it will impact directly the watchdog refresh timming.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	NULL_PTR

4.20 Reference WdgEcucPartitionRef

Maps the Wdg driver to zero or one ECUC partition to make the driver API available in this partition.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
	VARIANT-LINK-TIME: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.21 Container WdgSettingsConfig

WdgSettingsConfig

Configuration items for the different watchdog settings, including those for external watchdog hardware.

Included subcontainers:

- WdgExternalConfiguration
- WdgSettingsFast
- WdgSettingsSlow
- WdgSettingsOff

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.22 Parameter WdgDefaultMode

Wdg Default Mode

Default mode for watchdog driver initialization.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	WDGIF_SLOW_MODE
literals	['WDGIF_OFF_MODE', 'WDGIF_SLOW_MODE', 'WDGIF_FAST_MODE']

4.23 Parameter WdgErrorCounterLimit

Controls the Watchdog Error Counter Limit (FS_I_WD_CFG[WD_ERR_LIMIT]).

The Watchdog Error Counter is incremented by 2 after each bad Watchdog

refresh and decremented by 1 after each good Watchdog refresh.

When the Watchdog Error Counter reaches this limit, the Fault Error

Counter will be incremented by 1 and will trigger a Fail-Safe reaction

configured by FS_I_WD_CFG[WD_FS_IMPACT] (PmicWatchdogErrorImpact).

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	MAX_6
literals	['MAX_8', 'MAX_6', 'MAX_4', 'MAX_2']

4.24 Parameter WdgRefreshCounterLimit

Controls the Watchdog Refresh Counter Limit (FS_I_WD_CFG[WD_RFR_LIMIT]).

The Watchdog Refresh Counter is incremented by 1 after each good Watchdog refresh, but is reset to 0 every time there is a bad Watchdog refresh.

When the Watchdog Refresh Counter reaches this limit, and if the next Watchdog refresh is also good, the Fault Error Counter is decremented by 1.

Note: This field is editable only when the Watchdog Monitoring is enabled by OTP (i.e. PmicOtpFailSafeUnitConfiguration/PmicOtpSafetyIOConfiguration/ PmicWatchdogEnable = 'true')

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	MAX_6
literals	['MAX_6', 'MAX_4', 'MAX_2', 'MAX_1']

4.25 Parameter WdgErrorImpact

Configures the Fail-Safe reaction on RSTB and/or FS0B when the Watchdog Error Counter reaches its maximum value (FS_I_WD_CFG[WD_FS_IMPACT]).

Note: This field is editable only when the Watchdog Monitoring is enabled by OTP (i.e. PmicOtpFailSafeUnitConfiguration/PmicOtpSafetyIOConfiguration/ PmicWatchdogEnable = 'true')

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	FS0B__AND__RSTB__ASSERT
literals	['DISABLED', 'FS0B__ASSERT', 'FS0B__AND__RSTB__ASSERT']

4.26 Parameter WdgWindowPeriodEnable

This field is used to enable/disable the Watchdog Window Period (FS_WD__WINDOW__DUR[WD__WINDOW] = DISABLED).

The Watchdog Window Period is derived from the Fail-Safe oscillator with a +/-5% accuracy.

0 - Watchdog Window Period is Disabled. 1 - Watchdog Window Period is Enabled

Note: This field is editable only when the Watchdog Monitoring is enabled by OTP (i.e. PmicOtpFailSafeUnitConfiguration/PmicOtpSafetyIOConfiguration/PmicWatchdogEnable = 'true')

If this is disabled, WD__WINDOW will use default value (3ms) without any value in watchdog window period of Fast/Slow mode.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.27 Parameter WdgRecoveryWindowPeriod

Controls the Watchdog Window Duration when the device is in Fault Recovery Strategy (FS_WD_WINDOW_DUR[WDW_REC

When a fault is triggered by the MCU via its FCCU pins, the FS0B is asserted by the device and the watchdog window duration becomes automatically an open window (no more duty cycle), whose duration is given by this field.

The transition from the normal watchdog window duration to this new open window duration happens when the FCCU pin indicates an error and FS0B is asserted. If the MCU sends a good watchdog refresh before the end of the open window duration, the device switches bask to the normal watchdog window duration and associated duty cycle if the FCCU pins do not indicate an error anymore. Otherwise, a new open window period is started. If the MCU does not send a good watchdog refresh before the end of this second open window duration, then a reset pulse is generated and the Fail-Safe State Machine moves back to INIT_FS.

Note: This field is editable only when the Watchdog Monitoring is enabled by OTP (i.e. PmicOtpFailSafeUnitConfiguration/PmicOtpSafetyIOConfiguration/PmicWatchdogEnable = 'true')

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TIME_64MS
literals	['DISABLE', 'TIME_1MS', 'TIME_2MS', 'TIME_3MS', 'TIME_4MS', 'TIME_6MS', 'TIME_8MS', 'TIME_12MS', 'TIME_16MS', 'TIME_24MS', 'TIME_32MS', 'TIME_64MS', 'TIME_128MS', 'TIME_256MS', 'TIME_512MS', 'TIME_1024MS']

4.28 Parameter WdgServiceKeyValue

Configures the Watchdog Seed (FS_WD_SEED[WD_SEED]).

This value is used to calculate the value to be written in the WD_ANSWER

register during the OPEN watchdog window.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	23218
max	65534
min	1

4.29 Reference WdgPmicdeviceRef

Vendor specific: Wdg External Pmic device

Reference to the Pmicdevice configuration which set for the watchdog servicing routine implementation.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcucDefs/Pmic/PmicGlobalConfig/PmicDevice']

4.30 Reference WdgExternalTriggerCounterRef

Vendor specific:

Wdg External Trigger Counter

Reference to the GptChannel configuration which set for the watchdog servicing routine implementation.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcucDefs/Gpt/GptChannelConfigSet/GptChannelConfiguration']

4.31 Container WdgExternalConfiguration

WdgExternalConfiguration

Configuration items for an external watchdog hardware

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

4.32 Reference WdgExternalContainerRef

WdgExternalContainerRef

Reference to either

- a DioChannelGroup container in case the hardware watchdog is connected via DIO pins

- a SpiSequenceConfiguration container in case the watchdog hardware is accessed via SPI

Note:This parameter is not used by current implementation

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcucDefs/Dio/DioConfig/DioPort/DioChannelGroup', '/AUT↩ OSAR/EcucDefs/Spi/SpiDriver/SpiSequence']

4.33 Container WdgSettingsFast

WdgSettingsFast

Hardware dependent settings for the watchdog driver's fast mode.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.34 Parameter WdgWindowPeriod

Controls the Watchdog Window Period (FS_WD_WINDOW_DUR[WD_WINDOW]).

This new watchdog window period is effective after the next

watchdog refresh.

Note: Implementation Specific Parameter.

If the WdgWindowPeriodEnable is disabled and WdgDefaultMode is WDGIF_FAST_MODE mode, this value will not impact,

And WD_WINDOW will use default value (3ms) after FS_INIT phase is closed.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TIME_3MS
literals	['TIME_1MS', 'TIME_2MS', 'TIME_3MS', 'TIME_4MS', 'TIME_6MS', 'TIME_8MS', 'TIME_12MS', 'TIME_16MS', 'TIME_24MS', 'TIME_32MS', 'TIME_64MS', 'TIME_128MS', 'TIME_256MS', 'TIME_512MS', 'TIME_1024MS']

4.35 Parameter WdgClosedWindowDutyCycle

Controls the Watchdog Window Duty Cycle (FS_WD_WINDOW_DUR[WDW_DC]).

This new duty cycle is effective after the next watchdog refresh.

The first half of the window is said CLOSED and the second half is

said OPEN. A good watchdog refresh is a good watchdog answer during

the OPEN window. A bad watchdog refresh is a bad watchdog answer

during the OPEN window, no watchdog refresh during the OPEN window or a good watchdog answer during the CLOSED window. After a good

or bad watchdog refresh, a new window period starts immediately

for the MCU to keep the synchronization with the windowed watchdog.

Note: The OPEN window duty cycle is equal to (100 - PmicWatchdogClosedWindowDutyCycle)%.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DUTY_50_0
literals	['DUTY_68_75', 'DUTY_62_5', 'DUTY_50_0', 'DUTY_37_5', 'DUTY_31_25']

4.36 Container WdgSettingsSlow

WdgSettingsSlow

Hardware dependent settings for the watchdog driver’s slow mode.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.37 Parameter WdgWindowPeriod

Controls the Watchdog Window Period (FS_WD_WINDOW_DUR[WD_WINDOW]).

This new watchdog window period is effective after the next

watchdog refresh.

Note: Implementation Specific Parameter.

If the WdgWindowPeriodEnable is disabled and WdgDefaultMode is WDGIF_SLOW_MODE mode, this value will not impact,

And WD_WINDOW will use default value (3ms) after FS_INIT phase is closed.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	TIME_3MS
literals	['TIME_1MS', 'TIME_2MS', 'TIME_3MS', 'TIME_4MS', 'TIME_6MS', 'TI↩ ME_8MS', 'TIME_12MS', 'TIME_16MS', 'TIME_24MS', 'TIME_32MS', 'T↩ IME_64MS', 'TIME_128MS', 'TIME_256MS', 'TIME_512MS', 'TIME_1024↩ MS']

4.38 Parameter WdgClosedWindowDutyCycle

Controls the Watchdog Window Duty Cycle (FS_WD_WINDOW_DUR[WDW_DC]).

This new duty cycle is effective after the next watchdog refresh.

The first half of the window is said CLOSED and the second half is

said OPEN. A good watchdog refresh is a good watchdog answer during

the OPEN window. A bad watchdog refresh is a bad watchdog answer

during the OPEN window, no watchdog refresh during the OPEN window or a good watchdog answer during the CLOSED window. After a good

or bad watchdog refresh, a new window period starts immediately

for the MCU to keep the synchronization with the windowed watchdog.

Note: The OPEN window duty cycle is equal to (100 - PmicWatchdogClosedWindowDutyCycle)%.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DUTY_50_0
literals	['DUTY_68_75', 'DUTY_62_5', 'DUTY_50_0', 'DUTY_37_5', 'DUTY_31_25']

4.39 Container WdgSettingsOff

WdgSettingsOff

Hardware dependent settings for the watchdog driver's off mode.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.40 Parameter WdgWindowPeriod

Controls the Watchdog Window Period (FS_WD_WINDOW_DUR[WD_WINDOW]).

The watchdog window is disable (0 ms) after the next watchdog refresh.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DISABLE

4.41 Parameter WdgClosedWindowDutyCycle

Controls the Watchdog Window Duty Cycle (FS_WD_WINDOW_DUR[WDW_DC]).

This new duty cycle is effective after the next watchdog refresh.

The first half of the window is said CLOSED and the second half is

said OPEN. A good watchdog refresh is a good watchdog answer during

the OPEN window. A bad watchdog refresh is a bad watchdog answer

during the OPEN window, no watchdog refresh during the OPEN window

or a good watchdog answer during the CLOSED window. After a good

or bad watchdog refresh, a new window period starts immediately

for the MCU to keep the synchronization with the windowed watchdog.

Note: The OPEN window duty cycle is equal to (100 - PmicWatchdogClosedWindowDutyCycle)%.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	DUTY_50_0
literals	['DUTY_68_75', 'DUTY_62_5', 'DUTY_50_0', 'DUTY_37_5', 'DUTY_31_25']

4.42 Container WdgPublishedInformation

WdgPublishedInformation

Container holding all Wdg specific published information parameters

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.43 Parameter WdgTriggerMode

Wdg Trigger Mode

Watchdog trigger mode (toggle/window/both).

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	WDG_WINDOW
literals	['WDG_WINDOW']

4.44 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.45 Parameter ArReleaseMajorVersion

Vendor specific:

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.46 Parameter ArReleaseMinorVersion

Vendor specific:

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.47 Parameter ArReleaseRevisionVersion

Vendor specific:

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.48 Parameter ModuleId

Vendor specific:

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	102
max	102
min	102

4.49 Parameter SwMajorVersion

Vendor specific:

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.50 Parameter SwMinorVersion

Vendor specific:

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.51 Parameter SwPatchVersion

Vendor specific: Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	2
max	2
min	2

4.52 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows: <ModuleName>_>VendorId>_<VendorApiInfix>_ E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456_Write. This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	VR5510

4.53 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

This chapter describes the Tresos configuration plug-in for the Wdg_VR5510 Driver. The most of the parameters are described below.

Chapter 5

Module Index

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Module Documentation

6.1 Wdg_VR5510_HLD

6.1.1 Detailed Description

Data Structures

- struct Wdg_VR5510_DemConfigType

  DEM error reporting configuration. More...
- struct Wdg_43_VR5510_ConfigType

  Defines the configuration structure. More...

Types Reference

- typedef void(∗ Wdg_VR5510_CheckFaultStatusNotifyType) (const uint16 u16RegValue)

  This enumerated type contains the Wdg external driver's possible states.

Enum Reference

- enum Wdg_VR5510_ServiceIdType

  This enumerated type will contain the service ids for the watchodg functions.
- enum Wdg_VR5510_ErrorIdType

  Indicates the aditional det errors used by the watchdog driver.
- enum Wdg_VR5510_DeviceStatusType

  This enumerated type will contain the watchdog external driver's possible states.

Function Reference

- void Wdg_43_VR5510_Init (const Wdg_43_VR5510_ConfigType *pConfigPtr)
  This function initializes the WDG module.
- Std_ReturnType Wdg_43_VR5510_SetMode (WdgIf_ModeType Mode)
  Switches the watchdog into the mode Mode.
- void Wdg_43_VR5510_SetTriggerCondition (uint16 u16Timeout)
  Reset the watchdog timeout counter according to the timeout value passed.
- void Wdg_43_VR5510_Trigger (void)
  Triggers the Watchdog.

6.1.2 Data Structure Documentation

6.1.2.1 struct Wdg_VR5510_DemConfigType

DEM error reporting configuration.

This structure contains information DEM error reporting.

Definition at line 172 of file Wdg_43_VR5510.h.

6.1.2.2 struct Wdg_43_VR5510_ConfigType

Defines the configuration structure.

Definition at line 182 of file Wdg_43_VR5510.h.

Data Fields

- const WdgIf_ModeType Wdg_VR5510_DefaultMode
  The number of configured channels.
- const uint32 u16WdgPmicDeviceID
  ID of PMIC device.
- const Gpt_ChannelType Wdg_VR5510_TimerChannel
  Gpt Channel configured.
- const uint32 Wdg_VR5510_u32TriggerSourceClock
  The frequency of the configured timer channel.
- const Wdg_VR5510_DemConfigType * Wdg_VR5510_pDemConfig
  Pointer to Watchdog Specific implementation details.
- const Wdg_VR55xx_ConfigType * Wdg_VR5510_ModeSettings [3]
  Pointer to callout get the PMIC fault status by user for task notifications.

6.1.2.2.1 Field Documentation

6.1.2.2.1.1 Wdg_VR5510_DefaultMode `const WdgIf_ModeType Wdg_VR5510_DefaultMode`

The number of configured channels.

Definition at line 187 of file Wdg_43_VR5510.h.

6.1.2.2.1.2 u16WdgPmicDeviceID `const uint32 u16WdgPmicDeviceID`

ID of PMIC device.

Definition at line 191 of file Wdg_43_VR5510.h.

6.1.2.2.1.3 Wdg_VR5510_TimerChannel `const Gpt_ChannelType Wdg_VR5510_TimerChannel`

Gpt Channel configured.

Definition at line 195 of file Wdg_43_VR5510.h.

6.1.2.2.1.4 Wdg_VR5510_u32TriggerSourceClock `const uint32 Wdg_VR5510_u32TriggerSourceClock`

The frequency of the configured timer channel.

Definition at line 199 of file Wdg_43_VR5510.h.

6.1.2.2.1.5 Wdg_VR5510_pDemConfig `const Wdg_VR5510_DemConfigType* Wdg_VR5510_pDemConfig`

Pointer to Watchdog Specific implementation details.

< DEM error reporting configuration.

Definition at line 205 of file Wdg_43_VR5510.h.

6.1.2.2.1.6 Wdg_VR5510_ModeSettings `const Wdg_VR55xx_ConfigType* Wdg_VR5510_ModeSettings[3]`

Pointer to callout get the PMIC fault status by user for task notifications.

Definition at line 207 of file Wdg_43_VR5510.h.

6.1.3 Types Reference

6.1.3.1 Wdg_VR5510_CheckFaultStatusNotifyType

```
typedef void(* Wdg_VR5510_CheckFaultStatusNotifyType) (const uint16 u16RegValue)
```

This enumerated type contains the Wdg external driver's possible states.

This gives the call-out type for PMIC read status notifications.

Definition at line 165 of file Wdg_43_VR5510.h.

6.1.4 Enum Reference

6.1.4.1 Wdg_VR5510_ServiceIdType

```
enum Wdg_VR5510_ServiceIdType
```

This enumerated type will contain the service ids for the watchodg functions.

Precondition

To define WDG_GETVERSION_ID, WDG_VERSION_INFO_API has to be equal to STD_ON

Definition at line 103 of file Wdg_43_VR5510.h.

6.1.4.2 Wdg_VR5510_ErrorIdType

```
enum Wdg_VR5510_ErrorIdType
```

Indicates the aditional det errors used by the watchdog driver.

Definition at line 124 of file Wdg_43_VR5510.h.

6.1.4.3 Wdg_VR5510_DeviceStatusType

```
enum Wdg_VR5510_DeviceStatusType
```

This enumerated type will contain the watchdog external driver's possible states.

Enumerator

WDG_VR5510_DEVICE_UNINIT	The watchdog driver is not uninitialized.
WDG_VR5510_DEVICE_IDLE	= 0x01 The watchdog driver is currently idle, i.e not being triggered or mode changed
WDG_VR5510_DEVICE_BUSY	= 0x02 The watchdog driver is currently busy, i.e triggered or switched between modes
WDG_VR5510_DEVICE_INITIALIZING	= 0x03 The watchdog driver is currently initializing

Definition at line 144 of file Wdg_43_VR5510.h.

6.1.5 Function Reference

6.1.5.1 Wdg_43_VR5510_Init()

```
void Wdg_43_VR5510_Init (
            const Wdg_43_VR5510_ConfigType * pConfigPtr )
```

This function initializes the WDG module.

The `Wdg_43_VR5510_Init` function shall initialize the Wdg module and the watchdog hardware, i.e. it shall set the default watchdog mode and timeout period as provided in the configuration set.

Parameters

in	*ConfigPtr*	Pointer to configuration set.

6.1.5.2 Wdg_43_VR5510_SetMode()

```
Std_ReturnType Wdg_43_VR5510_SetMode (
            WdgIf_ModeType Mode )
```

Switches the watchdog into the mode Mode.

By choosing one of a limited number of statically configured settings (e.g. toggle or window watchdog, different timeout periods) the Wdg module and the watchdog hardware can be switched between the following three different watchdog modes using the `Wdg_43_VR5510_SetMode` function:

- WDGIF_OFF_MODE,
- WDGIF_SLOW_MODE.
- WDGIF_FAST_MODE.

Parameters

in	*Mode*	One of the following statically configured modes: 1. WDGIF_OFF_MODE, 2. WDGIF_SLOW_MODE, 3. WDGIF_FAST_MODE.

Returns

Std_ReturnType.

Return values

E_OK	Mode switch executed completely and successfully.
E_NOT_OK	The mode switch encountered errors.

6.1.5.3 Wdg_43_VR5510_SetTriggerCondition()

```
void Wdg_43_VR5510_SetTriggerCondition (
            uint16 u16Timeout )
```

Reset the watchdog timeout counter according to the timeout value passed.

Parameters

in	*Timeout*	value (milliseconds) for setting the trigger counter.

6.1.5.4 Wdg_43_VR5510_Trigger()

```
void Wdg_43_VR5510_Trigger (
            void )
```

Triggers the Watchdog.

The Wdg_Cbk_GptNotification shall trigger the hardware. It is set up as notification function for the Gpt timer that controls the trigger of the watchdog

Precondition

This API has to be set up as notification for the Gpt channels that is set up for Wdg

Parameters

in	*Wdg_InstanceID*	ID of Watchdog external device.

6.2 Wdg_VR5510_IPW

6.2.1 Detailed Description

Macros

- #define Wdg_VR5510_IPW_Init(Wdg_43_VR5510_IPW_ConfigPtr, Wdg_DeviceID)

  Mapping macro between Wdg_VR55xx initialization function and high level initialization function.
- #define Wdg_VR5510_IPW_SetMode(Wdg_43_VR5510_IPW_ConfigPtr, Wdg_DeviceID)

  Mapping macro between Wdg_VR55xx initialization function and high level set mode function.
- #define Wdg_VR5510_IPW_Trigger(Wdg_DeviceID)

  Mapping macro between Wdg_VR55xx trigger function and high level trigger function.
- #define Wdg_VR5510_IPW_CheckPmicState(Wdg_DeviceID)

  Mapping macro between Wdg_VR55xx check the PMIC state function and high level check Pmic's state function.
- #define Wdg_IPW_VR5510_ReadFsGFlagRegister(Wdg_DeviceID, FsGflagValue)

  Mapping macro between Wdg_VR55xx read the FS_G_GLAG register function and high level read FS_G_Flag function.
- #define Wdg_VR5510_IPW_ConfigType

  Wdg_VR5510_IPW_ConfigType.

6.2.2 Macro Definition Documentation

6.2.2.1 Wdg_VR5510_IPW_Init

```
#define Wdg_VR5510_IPW_Init(
            Wdg_43_VR5510_IPW_ConfigPtr,
            Wdg_DeviceID )
```

Mapping macro between Wdg_VR55xx initialization function and high level initialization function.

Definition at line 92 of file Wdg_43_VR5510_IPW.h.

6.2.2.2 Wdg_VR5510_IPW_SetMode

```
#define Wdg_VR5510_IPW_SetMode(
            Wdg_43_VR5510_IPW_ConfigPtr,
            Wdg_DeviceID )
```

Mapping macro between Wdg_VR55xx initialization function and high level set mode function.

Definition at line 97 of file Wdg_43_VR5510_IPW.h.

6.2.2.3 Wdg_VR5510_IPW_Trigger

```
#define Wdg_VR5510_IPW_Trigger(
              Wdg_DeviceID )
```

Mapping macro between Wdg_VR55xx trigger function and high level trigger function.

Definition at line 102 of file Wdg_43_VR5510_IPW.h.

6.2.2.4 Wdg_VR5510_IPW_CheckPmicState

```
#define Wdg_VR5510_IPW_CheckPmicState(
              Wdg_DeviceID )
```

Mapping macro between Wdg_VR55xx check the PMIC state function and high level check Pmic's state function.

Definition at line 107 of file Wdg_43_VR5510_IPW.h.

6.2.2.5 Wdg_IPW_VR5510_ReadFsGFlagRegister

```
#define Wdg_IPW_VR5510_ReadFsGFlagRegister(
              Wdg_DeviceID,
              FsGflagValue )
```

Mapping macro between Wdg_VR55xx read the FS_G_GLAG register function and high level read FS_G_Flag function.

Definition at line 112 of file Wdg_43_VR5510_IPW.h.

6.2.2.6 Wdg_VR5510_IPW_ConfigType

```
#define Wdg_VR5510_IPW_ConfigType
```

Wdg_VR5510_IPW_ConfigType.

Contains the information related to the hardware setup for SWT

Definition at line 94 of file Wdg_43_VR5510_IPW_Types.h.

6.3 Wdg_VR5510_IP

6.3.1 Detailed Description

Data Structures

- struct Wdg_VR55xx_ConfigType
  Wdg_VR55xx_ConfigType. More...

Enum Reference

- enum Wdg_VR55xx_PmicStateType
  This enumerated type will indicate the Pmic's failsafe state.

Function Reference

- Std_ReturnType Wdg_VR5510_VR55XX_Init (const Wdg_VR55xx_ConfigType ∗Vr55xx_pConfigPtr, const uint32 Wdg_DeviceID)
  Include Memory mapping specification.
- Std_ReturnType Wdg_VR5510_VR55XX_SetMode (const Wdg_VR55xx_ConfigType ∗Vr55xx_pConfigPtr, const uint32 Wdg_DeviceID)
  This function switches the mode for Wdg VR5510 module.
- void Wdg_VR5510_VR55XX_Trigger (const uint32 Wdg_DeviceID)
  This function triggers the external watchdog hardware.
- Wdg_VR55xx_PmicStateType Wdg_VR5510_VR55XX_PmicState (const uint32 Wdg_DeviceID)
  This function indicate the current Pmic's state.
- Std_ReturnType Wdg_VR5510_VR55XX_ReadFsGFlagRegister (const uint32 Wdg_DeviceID, uint16 ∗pFsGflagValue)
  This function returns the FS_G_FLAG register value.

6.3.2 Data Structure Documentation

6.3.2.1 struct Wdg_VR55xx_ConfigType

Wdg_VR55xx_ConfigType.

Contains the information related to the hardware setup for Wdg external device.

Definition at line 93 of file Wdg_VR5510_VR55XX_Types.h.

6.3.3 Enum Reference

6.3.3.1 Wdg_VR55xx_PmicStateType

```
enum Wdg_VR55xx_PmicStateType
```

This enumerated type will indicate the Pmic's failsafe state.

Precondition

To define WDG_GETVERSION_ID, WDG_VERSION_INFO_API has to be equal to STD_ON

Definition at line 108 of file Wdg_VR5510_VR55XX_Types.h.

6.3.4 Function Reference

6.3.4.1 Wdg_VR5510_VR55XX_Init()

```
Std_ReturnType Wdg_VR5510_VR55XX_Init (
            const Wdg_VR55xx_ConfigType * Vr55xx_pConfigPtr,
            const uint32 Wdg_DeviceID )
```

Include Memory mapping specification.

This function initializes the hardware of WDG module.

Sets up the values such as timeout value

Parameters

in	*Vr55xx_pConfigPtr*	Pointer to configuration set.
in	*Wdg_DeviceID*	Index of Pmic device.

Returns

Std_ReturnType

6.3.4.2 Wdg_VR5510_VR55XX_SetMode()

```
Std_ReturnType Wdg_VR5510_VR55XX_SetMode (
            const Wdg_VR55xx_ConfigType * Vr55xx_pConfigPtr,
            const uint32 Wdg_DeviceID )
```

This function switches the mode for Wdg VR5510 module.

Sets up the values such as timeout value

Parameters

in	*Vr55xx_pConfigPtr*	Pointer to configuration set.

Returns

Std_ReturnType

6.3.4.3 Wdg_VR5510_VR55XX_Trigger()

```
void Wdg_VR5510_VR55XX_Trigger (
            const uint32 Wdg_DeviceID )
```

This function triggers the external watchdog hardware.

Writes the trigger sequence on the hardware

Parameters

in	*Wdg_DeviceID*	ID of Pmic device.

Returns

void

Precondition

The Wdg module's environment shall make sure that the Wdg module has been initialized before the Wdg_↩ Trigger routine is called (Req: WDG104)

6.3.4.4 Wdg_VR5510_VR55XX_PmicState()

```
Wdg_VR55xx_PmicStateType Wdg_VR5510_VR55XX_PmicState (
            const uint32 Wdg_DeviceID )
```

This function indicate the current Pmic's state.

Writes the trigger sequence on the hardware

Parameters

in	*Wdg_DeviceID*	ID of Pmic device.

Returns

void

6.3.4.5 Wdg_VR5510_VR55XX_ReadFsGFlagRegister()

```
Std_ReturnType Wdg_VR5510_VR55XX_ReadFsGFlagRegister (
            const uint32 Wdg_DeviceID,
            uint16 * pFsGflagValue )
```

This function returns the FS_G_FLAG register value.

Read the FS_G_FLAG register value to use check fault error of PMIC device

Parameters

in	*Wdg_DeviceID*	ID of Pmic device.
out	*pFsGflagValue*	Pointer stores the FS_G_FLAG register value

Returns

void

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